

chapter 4

Non-linear equation:

the polynomials with degree > 1 .

it can be a series, polynomial, rational function, etc.

the function must be continuous on some interval

$$I = [a, b]$$

we want to solve $f(x) = 0$

solution of $f(x) = 0$ is called root of the function.

and denoted by x_* . It is also called zero of the function.

$f(x_*) = 0 \Rightarrow$ exact solution.

$(x - x_*)$ is a factor of $f(x)$.

How does the graph look like?

i) at $x = x_*$, the function will cross x -axis.
so the function changes sign.

ii) or it might just touch x -axis. Which implies first derivative changes sign at $x = x_*$.

here we will assume the graph will cross x -axis only once for $x \in I = [a, b]$

in other words factor $(x - x_*)$ has multiplicity 1.

We are looking for approximate solutions:

$f(x) \approx 0$, this also implies that $x - x_* \approx 0$.

What to do for the functions that

- With higher multiplicity

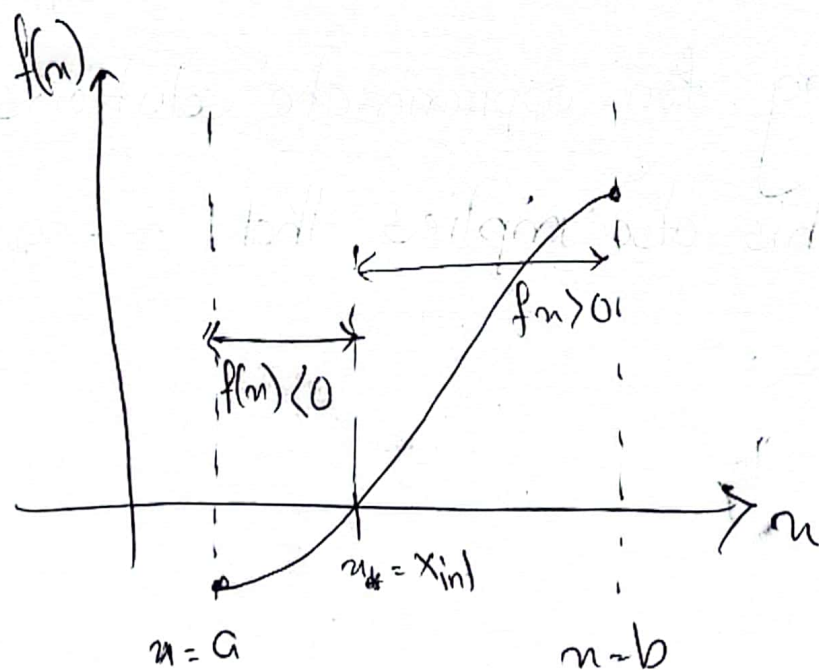
solution: we divide the domain of the function:

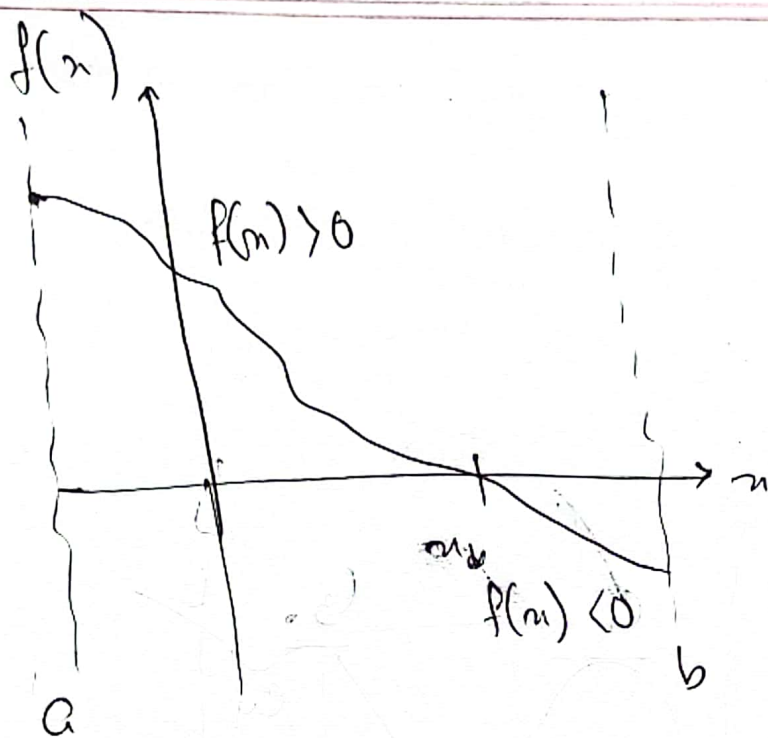
$$[a, b] = [a, c_1] \cup \dots \cup [c_n, b] = \bigcup_{i=1}^n [c_{i-1}, c_i]$$

$c_1, c_2, \dots$ are turning points.

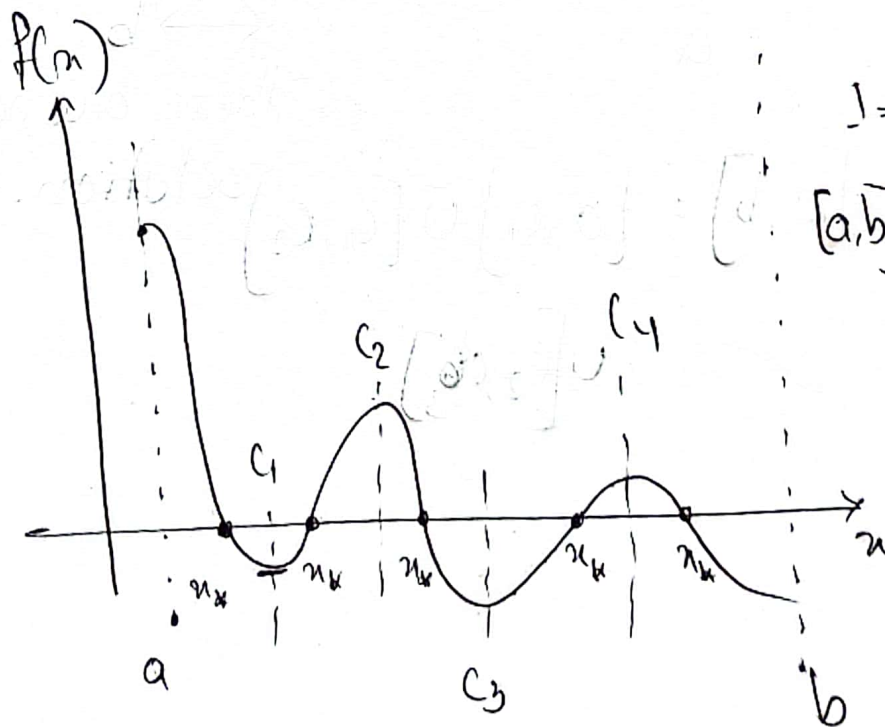
- ~~do~~ only touch x -axis

solution: we do the same as previous case.





multiple intercept:



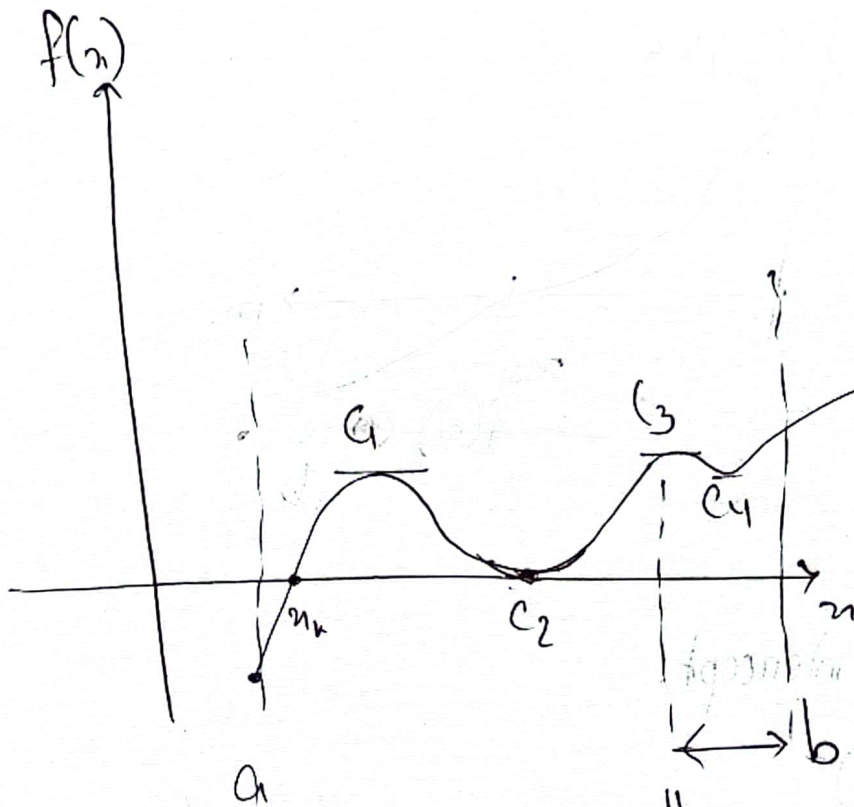
$$I = I_1 \cup I_2 \cup I_3 \cup I_4$$

$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup$$

$$[c_2, c_3] \cup [c_3, c_4]$$

$$[c_4, b]$$

just touch:



there are no
solution.

$$[a, b] = [a, c_1] \cup [c_1, c_2]$$

$$\cup [c_2, c_3]$$

Interval bisection method:

Intermediate value theorem:

If a function $f(x)$ is continuous on an Interval J
 $= [a, b]$ then for each $c \in J$, there exists a
real number x such that $f(x) = c$.

Now if $c = 0$, $f(x) = 0$ which is the root

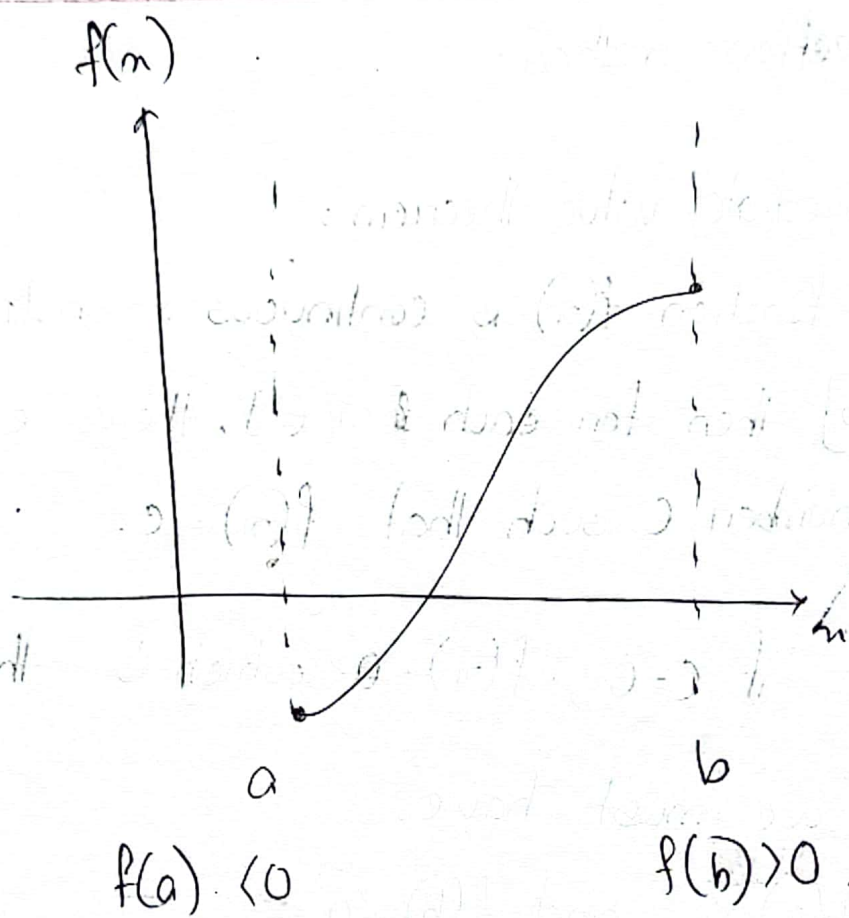
then we must have,

$f(a) > 0$ and $f(b) < 0$
or $f(a) < 0$ and $f(b) > 0$.

$\therefore f(a)f(b) < 0$

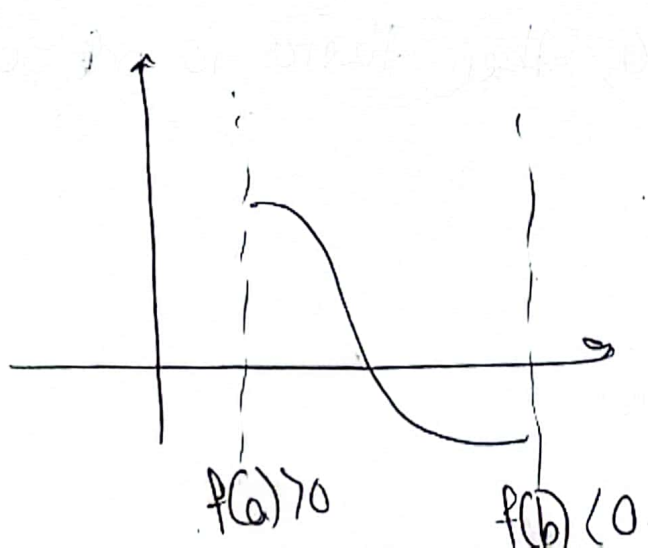
$\therefore$ if $f(a)f(b) > 0$, then there is no solution in
that interval.

converse is also true



$$\exists x_* \in [a, b]$$

$$f(a) f(b) < 0$$



$$\exists x_* \in [a, b]$$

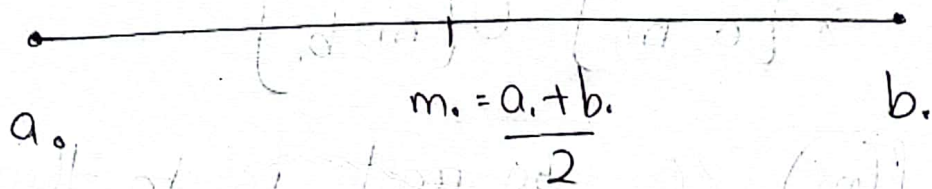
$$f(a) f(b) < 0$$

How does interval bisection method works:

let $I = [a_0, b_0]$, we also assume $f(a_0) > 0$ and $f(b_0) < 0$.

Now midpoint $m_0 = (a_0 + b_0) / 2$

$$[a_0, b_0] \rightarrow [a_0, m_0] \cup [m_0, b_0]$$



now, the root will fall in any of the sub-intervals

$$[a_0, m_0] \text{ or } [m_0, b_0]$$

$f(m_0) = 0$, the $x_r = m_0$.

$f(m_0) > 0$ root is on the right.

$$|m_1 - m_0| < \frac{1}{2} |b_0 - a_0| \quad k=0.$$

$$J_1 = [a, m_1] \cup$$

Now if error is within band, the m_0 is the root, else we continue.

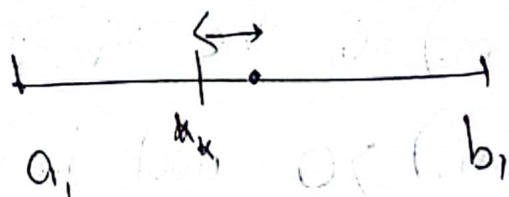
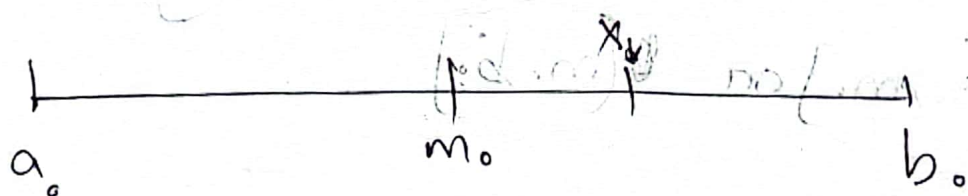
$$J_1 = [m_0, b_0]$$

$$= [a_1, b_1]$$

$$m_1 = \frac{1}{2}(b_1 + a_1)$$

$$J \rightarrow [a, m_1] \cup [m_1, b_1]$$

let $f(m_1) < 0$, so root is to the left



$$\text{error now: } |x_1 - m_1| \leq \frac{1}{2} |b_1 - a_1| = \frac{1}{2} |b_0 - a_0|$$

after n iteration,

$$|x_n - m_n| \leq \frac{|b_0 - a_0|}{2^{n+1}}$$

$$\frac{|b_0 - a_0|}{2^{n+1}} \leq \delta \Rightarrow n \geq \frac{\log(|b_0 - a_0|) - \log(\delta)}{\log(2)} - 1$$

if $a_0 = 1.5$, $b_0 = 3$ and $\delta = \epsilon_n = 1.1 \times 10^{-16}$

$$\text{then, } n \geq \frac{\log(3-1.5) - \log(1.1 \times 10^{-16})}{\log 2} - 1$$

$\therefore n \geq 53$ iteration.

Example:

Use interval bisection method to find solution accurate to within 10^{-3} for $f(x) = x^3 - 7x^2 + 14x - 6 = 0$, on interval $[1, 3.2]$

Solⁿ: $a_0 = 1$ $b_0 = 3.2$

$$f(1) = 1^3 - 7(1^2) + 14(1) - 6 = 2 > 0.$$

$$\text{and } f(3.2) = -0.11 < 0.$$

Since $f(a)f(b) < 0$,

$$\text{So } x_0 \in J_0 = [a_0, b_0]$$

$$m_0 = \frac{a_0 + b_0}{2} =$$

k	a_k	b_k	m_k	$f(a_k)$	$f(b_k)$	$f(m_k)$	n_k
0	0.1	3.2	2.1	2	-0.11	1.79	
1	2.1	3.2	2.65	1.79	-0.11	0.55	
2	2.65	3.2	2.925	0.55	-0.11	0.086	
3	2.925	3.2	3.0625	0.86	-0.11	-0.054	
4	2.925	3.0625					
...							
...							
...							
...							
9	2.998	3.002	3.000195	1.96×10^{-3}	-2.3×10^{-3}	-1.95×10^{-3}	

$< 10^{-3}$

Stopping condition

$$n_k = \underline{3.000}$$

three places as error bound 10^{-3} .

Fixed point iteration:

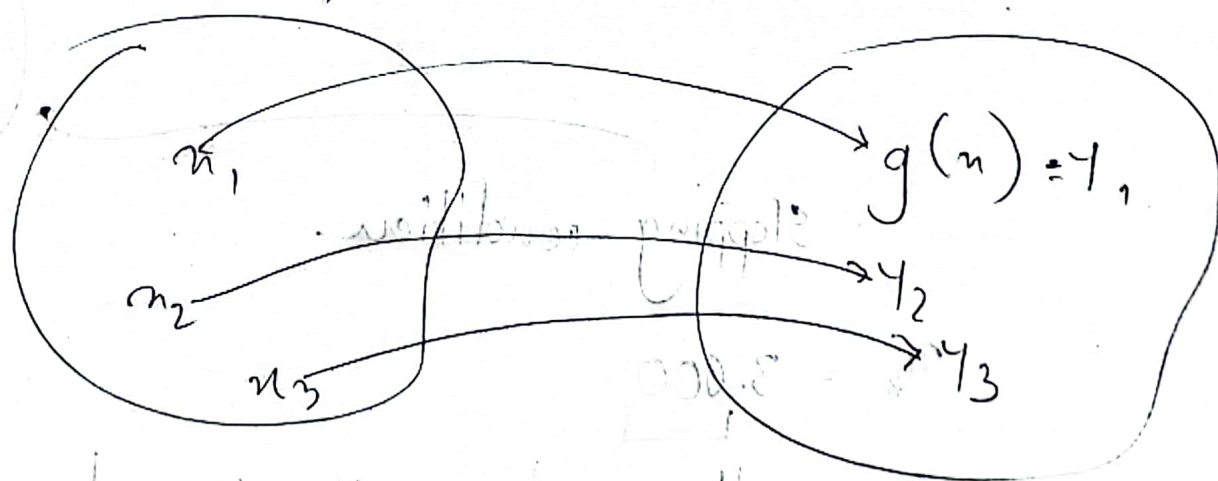
$$f(x) = 0 \longrightarrow g(x) - x = 0.$$

$f(x)$ and $g(x)$ must be continuous functions in some interval $I = [a, b]$

advantage: multiple roots can be taken care of.

$f(x) = 0$, if solution exists, this implies that $g(x) = x$.

this is fixed equation or mapping.

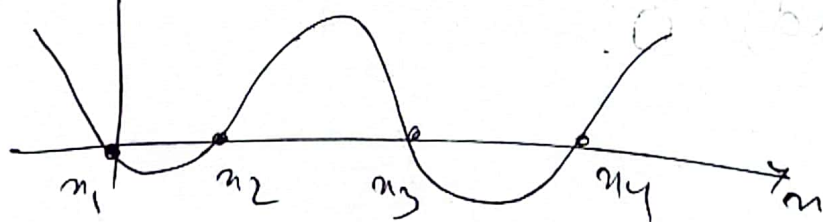


Now for non-trivial mapping $y = g(x) \neq x$.
 $\Rightarrow$ in trivial it could be.

So, Any point x that remains same under nontrivial mapping g is called, the fixed point under the mapping.

if $x_0 \in I \Rightarrow g(x_0) = x_0$, then we call x_0 a fixed point.

$f(x)$



$$\therefore f(x_1) = f(x_2) = f(x_3) \\ = f(x_4) = 0.$$

we want $g(x)$

i.e. $g(x_1) = x_1$

$g(x_2) = x_2$

$g(x_3) = x_3$

$g(x_4) = x_4$

fixed points

graph of $g(n) = n$

• It is clear that a line through the origin $(0,0)$ and (c,c) must have slope 1 (one).

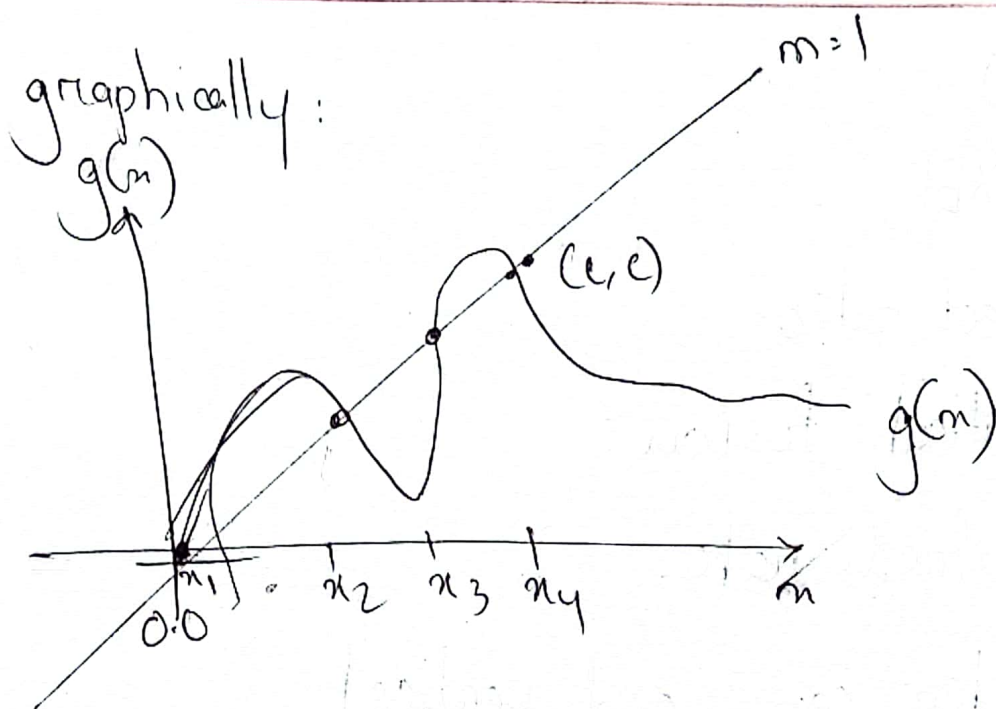
Therefore any graph of $g(n)$ vs n that intersect the line with slope one must have a fixed point.

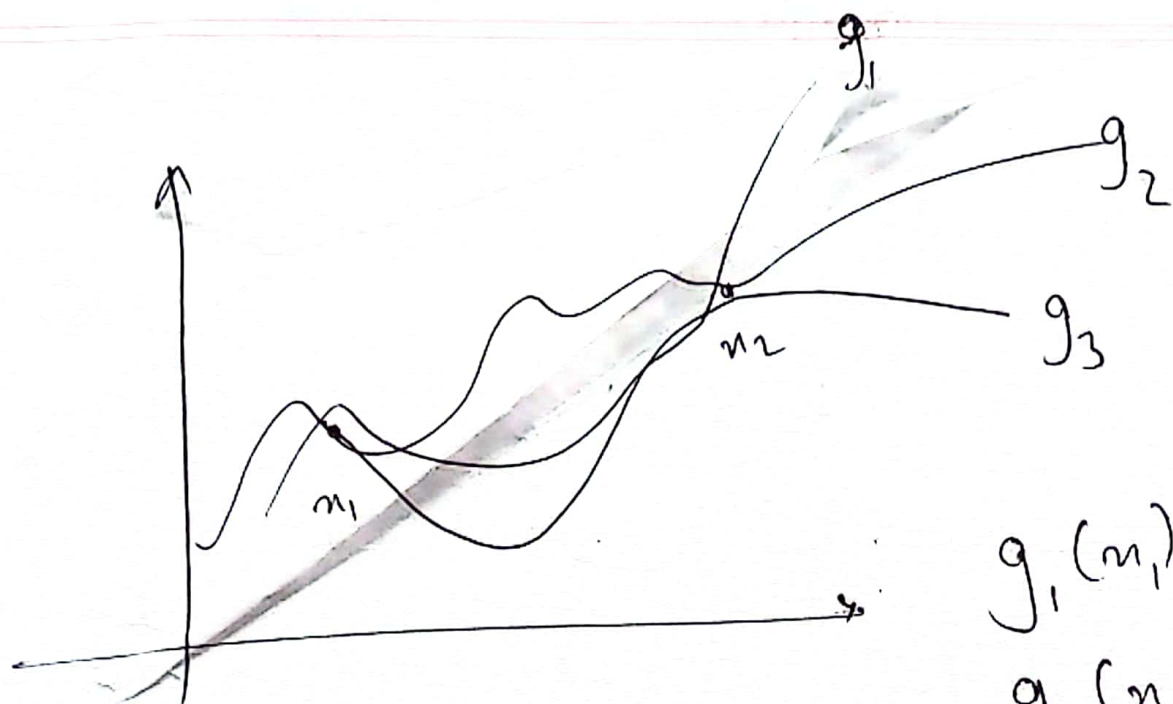
if $g(n)$ intersect $n = d$, then $g(d) = d$

$\therefore n = d$ is a fixed point of $g(n)$

$\therefore n = d$ is a root of the fun^c.

$$f(d) = 0.$$





$$g_1(n_1) = n_1$$

$$g_2(n_1) = n_1$$

$$g_3(n_1) = n_1$$

$$g_1(n_2) = n_2$$

$$g_2(n_2) = n_2$$

$$g_3(n_2) = n_2$$

$$g(n) = \frac{n+2}{2}$$

$$g(0) = 1$$

$\Rightarrow n=0$ is not fixed point

$$g(1) = 1.5$$

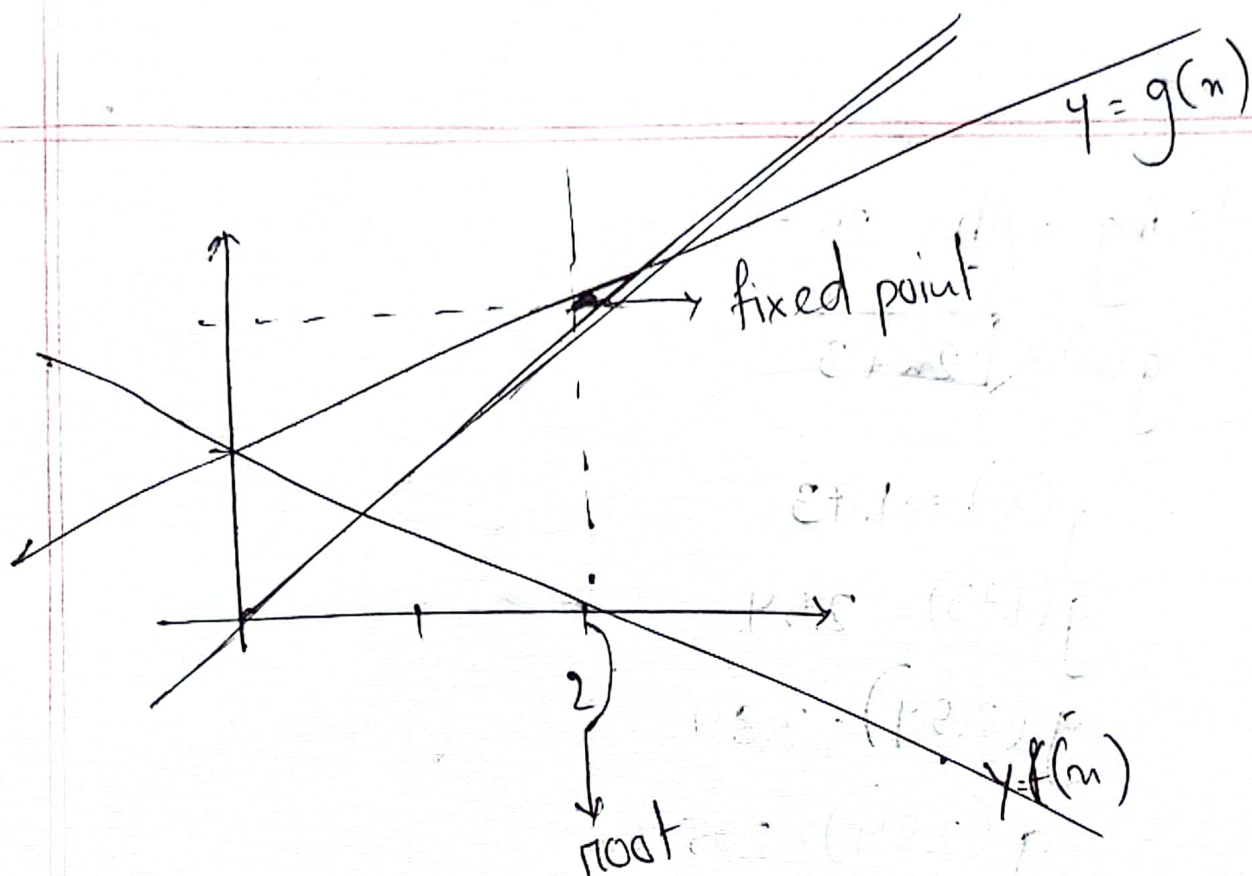
$\Rightarrow n=1.5$ " " " "

$$g(2) = 2$$

$\Rightarrow n=2$ is a fixed point

$$\therefore f(n) = g(n) - n = \frac{n+2}{2} - n = -\frac{1}{2}n + 1$$

has a root at 2.



$$f(n) = n^2 - 2n - 3 = 0.$$

in reality $n^2 - 2n - 3 = (n-3)(n+1)$, $f(n)$ has two roots at $n = 3, -1$

lets try out $g(n)$

$$1) n^2 - 2n - 3 = 0 \Rightarrow n = \sqrt{2n+3}$$

$$2) n^2 - 2n - 3 = n(n-2) - 3 \Rightarrow n = \frac{3}{n-2}$$

$$3) n^2 - 2n - 3 = 0 \Rightarrow n = n^2 - n - 3$$

$$4) n^2 - 2n - 3 = 0 \Rightarrow 2n^2 - 2n = n^2 + 3 \Rightarrow n = \frac{n^2 + 3}{2n - 2}$$

starting with $n = 0$

$$g(n) = \sqrt{2n+3}$$

$$g(0) = 1.73$$

$$g(1.73) = 2.54$$

$$g(2.54) = 2.84$$

$$g(2.84) = 2.95$$

$$g(2.95) = 2.98$$

$$g(2.98) = 2.99$$

$$g(2.99) = 3.00$$

$$g(n) = n^2 - n - 3 \quad \rightarrow \text{not convergent}$$

$$g(0) = -300$$

$$g(-3) = 9.00$$

$$g(9) = 69$$

$$g(69) = 4.69 \times 10^3$$

$$g(n) = \frac{n^2 + 3}{2n - 2}$$

$$g(0) = -1.5105$$

$$g(-1.5) = -1.05$$

$$g(-1.05) = -1.00$$

question is what $g(n)$ to take, what n to take.

answer $\rightarrow$ "Contraction Mapping theorem"

$$g(n) = \sqrt{2n+3}$$

$$g(42) = 9.33$$

$$g(0) = 1.73$$

$$g(9.33) = 4.65$$

$$g(1.73) = 2.54$$

$$g(4.65) = 3.51$$

$$g(2.54) = 2.84$$

$$g(3.51) = 3.17$$

$$g(2.84) = 2.95$$

$$g(3.17) = 3.06$$

$$g(2.95) = 2.98$$

$$g(3.06) = 3.02$$

$$g(2.98) = 2.99$$

$$g(3.02) = 3.01$$

$$g(2.99) = 3.00$$

$$g(3.01) = 3.00$$

$$g(3.00) = 3$$

Both $n_0 = 0$ & $n_0 = 42$ converges to root, $n_* = 3$

Even though $n_0 = 0$ is closer to $n_* = -1$, it

converges to $n_* = 3$

$$g(n) = n^2 - n - 3$$

$$g(42) = 1.72 \times 10^3$$

$$g(1.72 \times 10^3) = 2.95 \times 10^6$$

$$g(2.95 \times 10^6) = 8.72 \times 10^{12}$$

$$g(0) = -3$$

$$g(-3) = 9$$

$$g(9) = 69$$

$\therefore$ both diverges

$$g(n) = \frac{n^2 + 3}{2n - 2}$$

$$g(42) = 21.6$$

$$g(21.6) = 11.4$$

$$g(11.4) = 6.39$$

$$g(6.39) = 4.07$$

$$g(4.07) = 3.19$$

$$g(3.19) = 3.01$$

$$g(3.01) = 3.00$$

$$g(0) = -1.5$$

$$g(-1.5) = -1.05$$

$$g(-1.05) = -1$$

$$g(-1) = -1$$

$\therefore n_0 = 0$ converges to nearest root -1

$$\lambda = \left| \frac{\Delta g(n_k)}{\Delta n} \right| = \left| \frac{g(n_{k+1}) - g(n_k)}{n_{k+1} - n_k} \right| < 1$$

Contraction mapping theorem:

If g is a contraction mapping on $L = [a, b]$ then

1) There exists a unique fixed point $n_* \in L$
with $g(n_*) = n_*$

2) For any $n_0 \in L$, the iteration $n_{k+1} = g(n_k)$
will converge to n_* as $k \rightarrow \infty$

$$\lambda = |g'(\text{root})|$$

000

$$g(x) = \sqrt{2x+3} = (2x+3)^{1/2}$$

$$g'(x) = \frac{1}{2}(2x+3)^{-1/2}$$

$$\lambda = |g'(1)| = 1 \quad (\text{not } < 1)$$

$$\lambda = |g'(3)| = \frac{1}{3} (< 1)$$

$\therefore$ both $x_1 = 0, x_2 = 42$, converge to $x_* = 3$
~~but $x_1 = 0, x_2 = 42$ converge to~~

$\lambda = 0 \rightarrow$ super linear convergence

$\rightarrow$ fastest convergence
 less iteration

$0 < \lambda < 1 \rightarrow$ linear convergence required

$\circ$ $\rightarrow$ slowly will converge.

$\lambda = 1$ ignore

$\lambda > 1 \rightarrow$ Diverge

$$g(x) = x^2 - x - 3$$

$$g'(x) = 2x - 1$$

$$\lambda = |g'(-1)| = |-1| = 1 \quad (\text{not } < 1)$$

$$\lambda = |g'(3)| = |5| = 5 \quad (\text{not } < 1)$$

$\therefore$ diverge.

$$g(x) = \frac{x^2 + 3}{2x - 2}$$

$$\frac{dg}{dx} = \frac{x^2 - 2x - 3}{2(x-1)^2}$$

$$\lambda = |g'(-1)| = 0$$

$$\lambda = |g'(3)| = 0$$

$$\therefore x_1 = 0 \rightarrow -1$$

$$x_2 = 42 \rightarrow 3$$

Example:

$$f(n) = n^3 - 2n^2 - n + 2.$$

a) state the actual root of $f(n)$

b) construct 3 different ~~to~~ $g(n)$ such that $f(n) = 0$.

c) find the converging rate $g(n)$ and what it will converge to.

$$a) f(n) = n^3 - 2n^2 - n + 2 = 0.$$

$$n^2(n-2) - 1(n-2) = 0.$$

$$(n^2 - 1)(n - 2) = 0.$$

$$n = \begin{matrix} 1 \\ -1 \\ 2 \end{matrix}$$

$$b) i) x^3 - 2x^2 - x + 2 = 0.$$

$$x = \frac{x^3 - 2x^2 + 2}{x}$$

$$ii) x^3 - 2x^2 - x + 2 = 0.$$

$$x(x^2 - 2x - 1) = -2$$

$$x = \frac{-2}{x^2 - 2x - 1}$$

$$iii) x^3 - 2x^2 - x + 2 = 0.$$

$$2x^2 = x^3 - x + 2$$

$$x = \frac{1}{\sqrt{2}} \sqrt{x^3 - x + 2}$$

$$c) \Downarrow g(n) = n^3 - 2n^2 + 2$$

$$g'(n) = 3n^2 - 4n$$

$$\lambda = |g'(n_k)| = \begin{matrix} 7 \rightarrow n_k = -1 \\ 1 \rightarrow n_k = 0 \\ 4 \rightarrow n_k = 2 \end{matrix}$$

all divergent

$$ii) g(n) = \frac{-2}{n^2 - 2n - 1}$$

$$g'(n) = \frac{4(n-1)}{(n^2 - 2n - 1)^2}$$

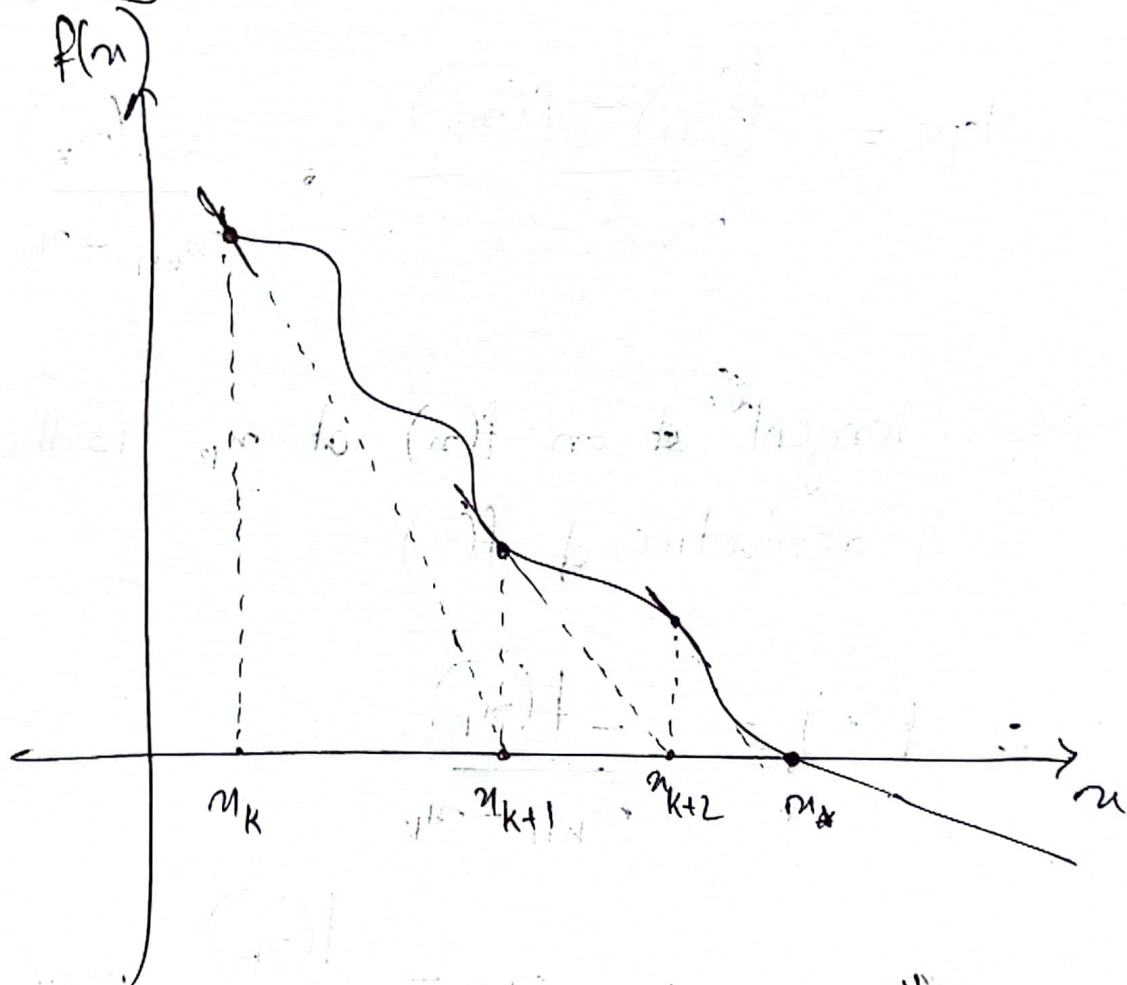
$$\lambda = g'(n_k) = \begin{matrix} 2 & n_k = -1 \\ 0 & n_k = 1 \end{matrix}$$

$\therefore$ only converge to $n_k = 1$

Newton's method:

Also called Newton Raphson method

method of finding $g(n)$ is different
choosing n . is also different



$n_* - n_k$ is getting smaller.

$$n_* - n_k \approx 0 \text{ as } k \rightarrow \infty$$

According to the discuss on the previous case, it is clear that x_k and x_{k+1} are both on same tangent line.

$$\text{slope} = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k} = \frac{-f(x_k)}{x_{k+1} - x_k}$$

Now, tangent ~~at~~ on $f(x)$ at x_k is the first derivative of $f(x)$

$$\therefore f'(x) = \frac{-f(x_k)}{x_{k+1} - x_k}$$


$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \approx g(x_k)$$

$$g'(n) = \frac{d}{dn} \left(n - \frac{f(n)}{f'(n)} \right)$$

$$= 1 - \frac{f'(n)f'(n) - f(n)f''(n)}{(f'(n))^2}$$

$$= \frac{f(n)f''(n)}{(f'(n))^2}$$

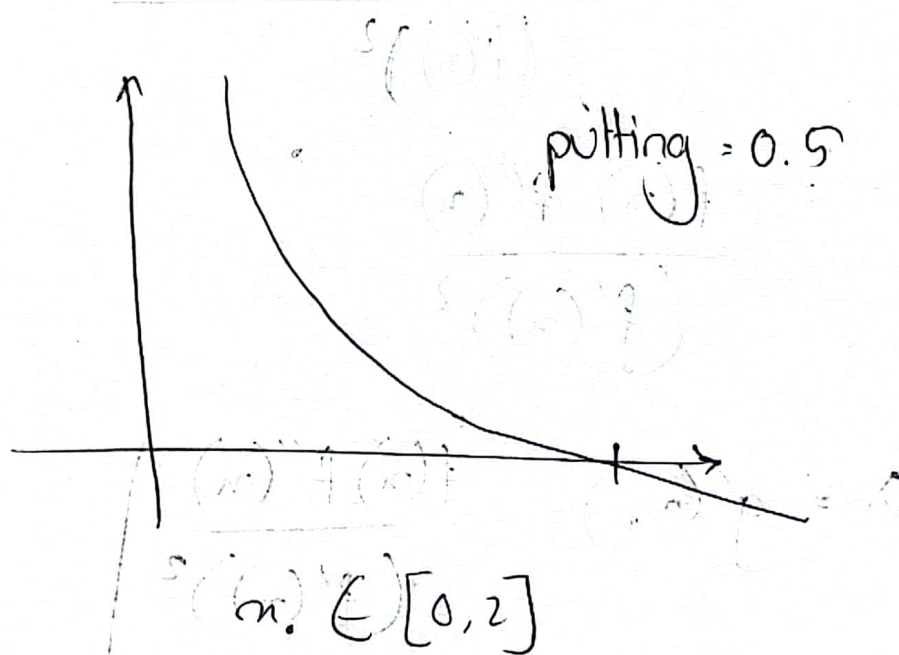
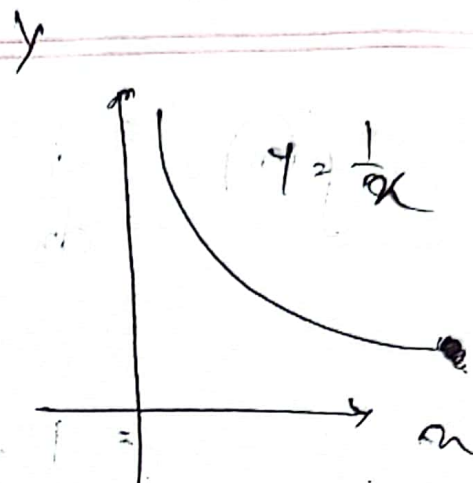
$$\lambda = g'(n_*) = \left. \frac{f(n)f''(n)}{(f'(n))^2} \right|_{n=n_*}$$

$$= \frac{\overset{0}{\cancel{f(n_*)}} f''(n_*)}{(f'(n_*))^2} = 0.$$

$\therefore$ Newton's method is super-linear.

Example: $f(x) = \frac{1}{x} - a$

$\therefore m_x = \frac{1}{a}$



$\therefore \frac{(x-1)}{(x-1)}$

$$n_{k+1} = n_k - \frac{f(n_k)}{f'(n_k)}$$

$$n_k = \frac{\frac{1}{n_k} - 0.5}{\frac{1}{n_k^2}}$$

$$n_k = \frac{1 - 0.5n_k}{n_k}$$

$$n_k + n_k - 0.5n_k^2 = 2n_k - 0.5n_k^2$$

$$= n_k - \frac{\frac{1}{n_k} - 0.5}{\frac{d}{dn} \left(\frac{1}{n_k} - 0.5 \right)}$$

$$\therefore n_{k+1} = 2n_k - 0.5n_k^2$$

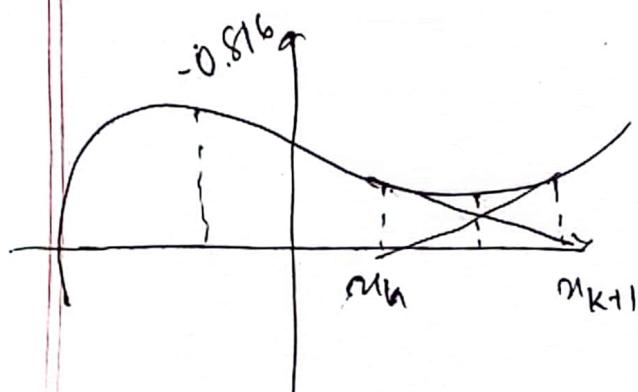
Let's start the iteration at $n_0 = 1$



k	x_k	x_{k+1}
0	1	1.5
1	1.5	1.875
2	1.875	
3	1.9921875	
4	1.999969482	
5		
6		

k	x
0	1
1	1.5
2	1.875
3	1.9921875
4	1.999969482
5	2
6	2

When Newton's method do not work:
when there is a turning point



If $f(x) = x^3 - 2x + 2$
Finding turning point:
 $f'(x) = 3x^2 - 2 = 0$

$$x = \pm \sqrt{2/3}$$

for marks since $x < -0.816$.

Aitke Acceleration:

In this method we learn how to accelerate the convergence rate for a linear convergent function.

$\Rightarrow$ for $\lambda = \text{constant} \neq 0$.

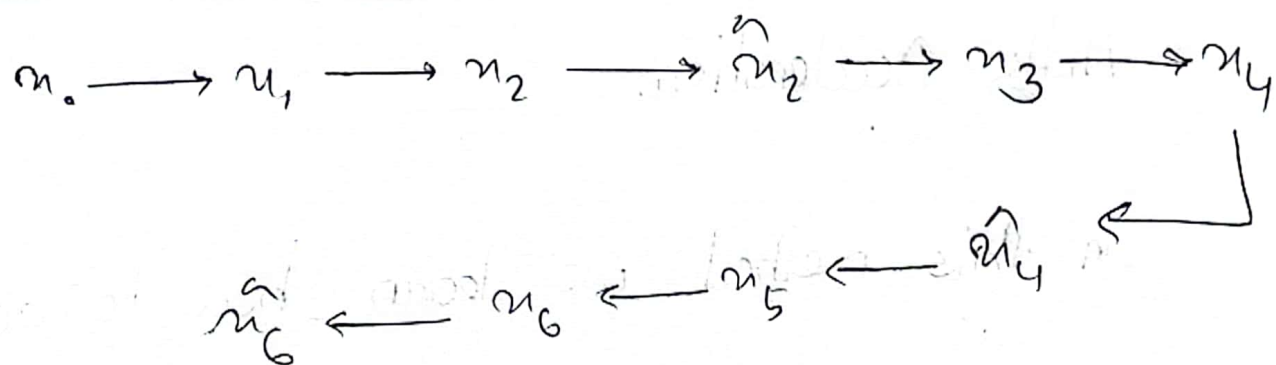
$$\cancel{x_{k+1}} = \cancel{\lambda x_k}$$

iteration formula:

$$x_{k+2} = \frac{x_k - (x_{k+1} - x_k)^2}{x_{k+2} - 2x_{k+1} + x_k}$$

accelerated value $\nearrow$

for every 2 iteration - acceleration occurs



Example: $f(n) = \frac{1}{n} - 0.5$ [n_k is at 2]

construct $g(n)$ such that $n=2$ is
a fixed point of $g(n)$.

$$\text{let } g(n) = n + \frac{1}{16} f(n) = n + \frac{1}{16} \left(\frac{1}{n} - 0.5 \right)$$

valid since $g(2) = 2$.

$$\lambda = g'(n_k) = 1 + \frac{1}{16} \left(-\frac{1}{n^2} \right)$$

$$= 0.984375 < 1$$

linear convergence but very slow.

— lets start with $x_0 = 1.5$.

— we consider upto 7 significant values.

$$x_0 = 1.5$$

$$x_1 = g(x_0) = 1.510417$$

$$0.489583 \rightarrow (x_1 - x_*)$$

$$x_2 = g(x_1) = 1.520546$$

$$0.479454 \rightarrow (x_2 - x_*)$$

$$x_3 = g(x_2) = 1.530400$$

$$\vdots$$
$$x_{818} = g(x_{817}) = 1.999999$$

$$0.000000$$

$$\rightarrow (x_{818} - x_*)$$

Aitken acceleration:

$$n_0 = 1.5$$

$$n_1 = g(n_0) = 1.510417$$

$$(n_1 - n_*) = 0.0189583$$

$$n_2 = g(n_1) = 1.520546$$

$$(n_2 - n_*) = 0.009454$$

$$\hat{n}_2 = n_0 - \frac{(n_1 - n_0)^2}{n_2 - 2n_1 + n_0} = 1.877604$$

$$(\hat{n}_2 - n_*) = 0.12239$$

$$n_3 = g(\hat{n}_2) = 1.879641$$

$$(n_3 - n_*) = 0.12035$$

$$n_4 = g(n_3) = 1.881642$$

$$(n_4 - n_*) = 0.007366$$

$$\hat{n}_4 = \hat{n}_2 - \frac{(n_3 - \hat{n}_2)^2}{2n_4 - 2n_3 + \hat{n}_2} = 1.992364$$

$$(\hat{n}_4 - n_*) = 0.007366$$

$$\hat{u}_8 = g(m_7) = 2.0000000$$

$$(\hat{u}_8 - u_*) \leq 0.0000000$$

Secant method

Also called Quasi Newton Method

$$x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$$

Example: $f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-3}$

$$\left. \begin{array}{l} x_0 = 0.003 \\ x_{-1} = 0.002 \end{array} \right\} \text{ will always be given.}$$

Show 3 iterations.

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)(x_0 - x_{-1})}{f(x_0) - f(x_{-1})} \\ &= 0.03 - \frac{f(0.03)(0.03 - 0.02)}{f(0.03) - f(0.02)} \end{aligned}$$

H.W.

Example: $f(x) = \frac{1}{x} - 0.5$ $x_0 = 0.25, x_1 = 0.5$

$$x_{k+1} = x_k - \frac{\left(\frac{1}{x_k} - 0.5\right) (x_k - x_{k-1})}{\left(\frac{1}{x_k} - 0.5\right) - \left(\frac{1}{x_{k-1}} - 0.5\right)}$$

k	x_k
0	0.25
1	0.5
2	0.6875
3	1.01562
4	1.3540
5	1.68205
6	1.8973
7	1.98367
⋮	
12	2.0000